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Course: Data Structures – Lab 2
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Task 3:

//25K0538, Muhammad Farhan

//For this task, the most efficient and suitable way to store is to use Jagged Array.

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    string departments[4] = {"SE", "AI", "CS", "DS"};
```

```
    int sizes[4] = {3,4,2,1};
```

```
    double** GPA = new double*[4]; //4 departments
```

```
    GPA[0] = new double[3] {3.3, 3.67, 3}; //SE
```

```
    GPA[1] = new double[4] {2.3, 2.67, 2, 3}; //AI
```

```
    GPA[2] = new double[2] {4, 3.3}; //CS
```

```
    GPA[3] = new double[1] {2.67}; //DS
```

```
    for (int i = 0; i < 4; i++)
```

```
    {
```

```
        cout << departments[i] << ": ";
```

```
        for (int j = 0; j < sizes[i]; j++)
```

```
        {
```

```
            cout << GPA[i][j] << " ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    for (int i = 0; i < 4; i++)
```

```
    {
```

```
        delete[] GPA[i];
```

```
    }
```

```
    delete[] GPA;
```

```
    return 0;
```

```
}
```

```
SE: 3.3 3.67 3
AI: 2.3 2.67 2 3
CS: 4 3.3
DS: 2.67
```

Task 4:

//25K0538, Muhammad Farhan

//For this task, the most efficient and suitable way to store is to use Jagged Array.

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int rows[4] = {3,2,1,1}; //size is small so its easier to input names
```

```
    string** Names = new string*[4]; //4 rows
```

```
    Names[0] = new string[3] ; //R1
```

```
    Names[1] = new string[4] ; //R2
```

```
    Names[2] = new string[1] ; //R3
```

```
    Names[3] = new string[1] ; //R4
```

```
    //taking inputs
```

```
    for (int i = 0; i < 4; i++)
```

```
    {
```

```
        cout << "\nRow " << i+1 << ": " << endl;
```

```
        for (int j = 0; j < rows[i]; j++)
```

```
        {
```

```
            cout << "Enter name for R" << i+1 << "C" << j+1 << ": ";
```

```
            cin >> Names[i][j];
```

```
            cout << endl;
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    //showing output
```

```
    for (int i = 0; i < 4; i++)
```

```
    {
```

```
        cout << "Row " << rows[i] << ": ";
```

```
        for (int j = 0; j < rows[i]; j++)
```

```
        {
```

```
            cout << Names[i][j] << " ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    for (int i = 0; i < 4; i++)
```

```
    {
```

```
        delete[] Names[i];
```

```
    }
```

```
    delete[] Names;
```

```
    return 0;
```

```
}
```

Row 1:
Enter name for R1C1: Farhan

Enter name for R1C2: Faraz

Enter name for R1C3: Ahmed

Row 2:
Enter name for R2C1: Jawwad

Enter name for R2C2: Rehan

Row 3:
Enter name for R3C1: Abbas

Row 4:
Enter name for R4C1: Aqsa

Row 3: Farhan Faraz Ahmed
Row 2: Jawwad Rehan
Row 1: Abbas
Row 1: Aqsa

Task 5:

//25K0538, Muhammad Farhan

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int sizes[4] = {3,4,2,1};
```

```
    int** array = new int*[4];
```

```
    array[0] = new int[3] {3, 4, 3};
```

```
    array[1] = new int[4] {10, 2, 7, 3};
```

```
    array[2] = new int[2] {4, 8};
```

```
    array[3] = new int[1] {9};
```

```
    int max = 0;
```

```
    int max_index = 0;
```

```
    for (int i = 0; i < 4; i++)
```

```
    {
```

```
        int count = 0;
```

```
        for (int j = 0; j < sizes[i]; j++)
```

```
        {
```

```
            count++;
```

```
        }
```

```
        if (count > max)
```

```
        {
```

```
            max = count;
```

```
            max_index = i;
```

```
        }
```

```
    }
```

```
    cout << "Row " << max_index+1 << " contains the maximum number of elements." << endl;
```

```
    cout << "Size: " << max << endl;
```

```
    for (int i = 0; i < max; i++)
```

```
    {
```

```
        cout << array[max_index][i] << " ";
```

```
    }
```

```
    for (int i = 0; i < 4; i++)
```

```
    {
```

```
        delete[] array[i];
```

```
    }
```

```
    delete[] array;
```

```
    return 0;
```

```
}
```

```
Row 2 contains the maximum number of elements.  
Size: 4  
10 2 7 3 ↵
```